



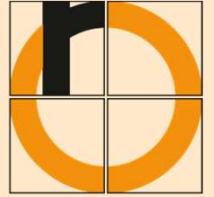
Systems Engineering - SySe

Chapter 4 – Product Life Cycle

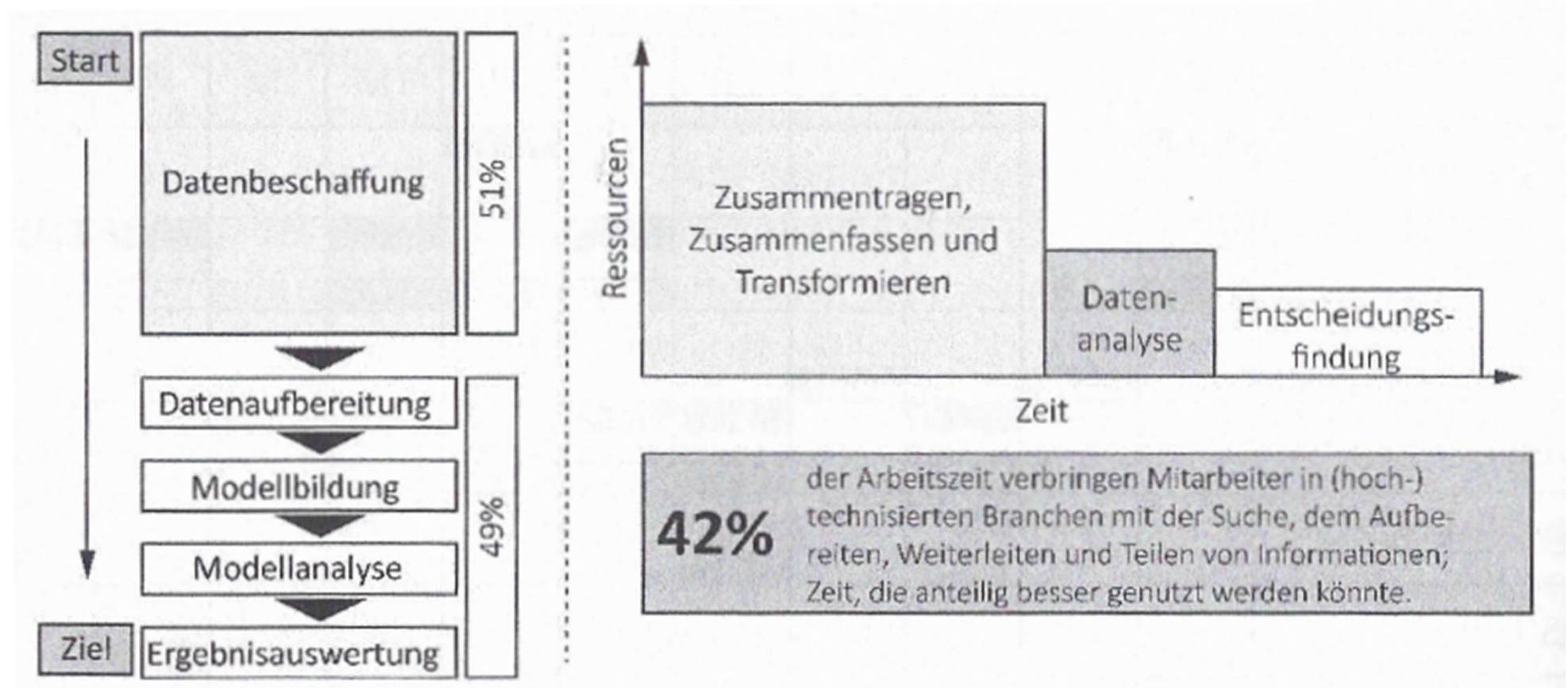
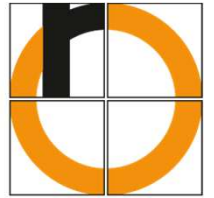
Prof. Dr. Silke Lechner-Greite

Content

- Product Life Cycle
- Product Life Cycle Stages



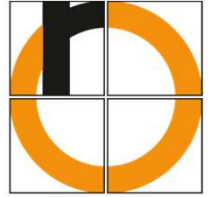
What do engineers spend most of their working time on?



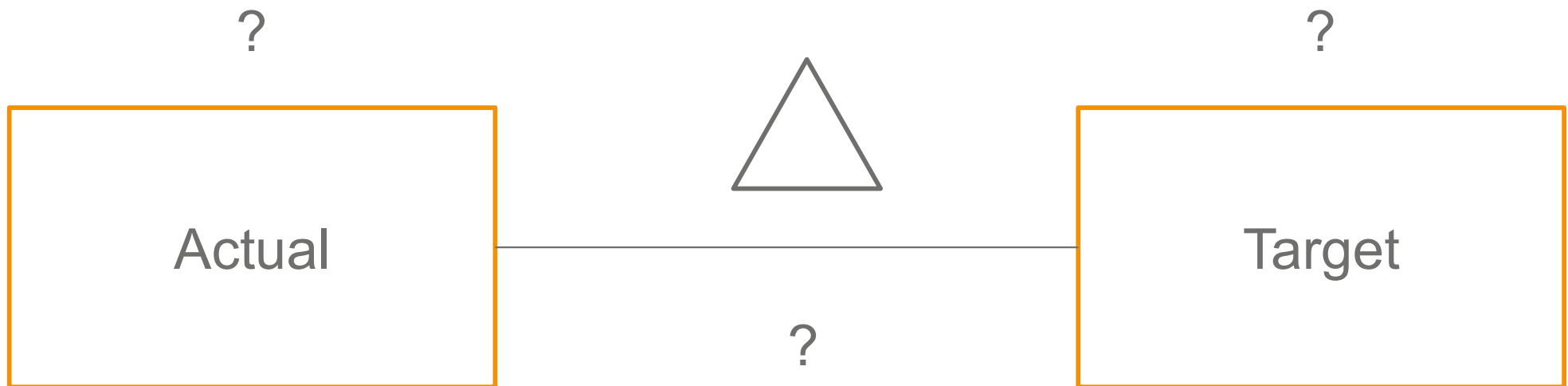
Source: GfSE Book: Figure 15, page 66: Engineers like searching...

Recap:

What are important components in SE?



- Problem as difference between ACTUAL and TARGET

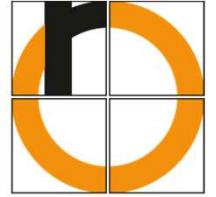


“ ... if you want to solve a problem, there are usually many factors that determine it ... ”

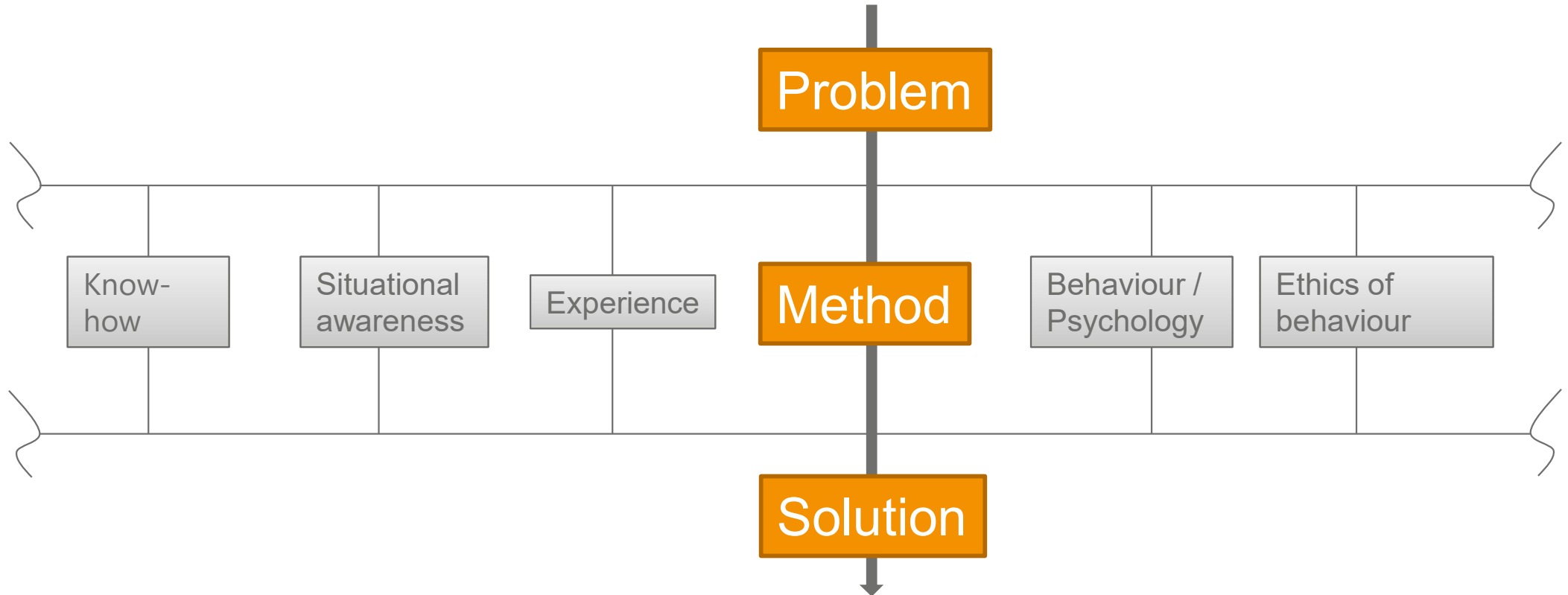
(1) Source: Haberfellner et al., Systems Engineering Grundlagen und Anwendung, orell füssli 2018 (14. Auflage)

Recap:

What are important components in SE?



- System Engineering as a methodological component in problem solving

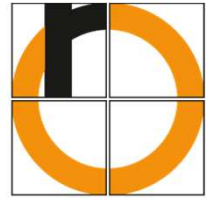


But:

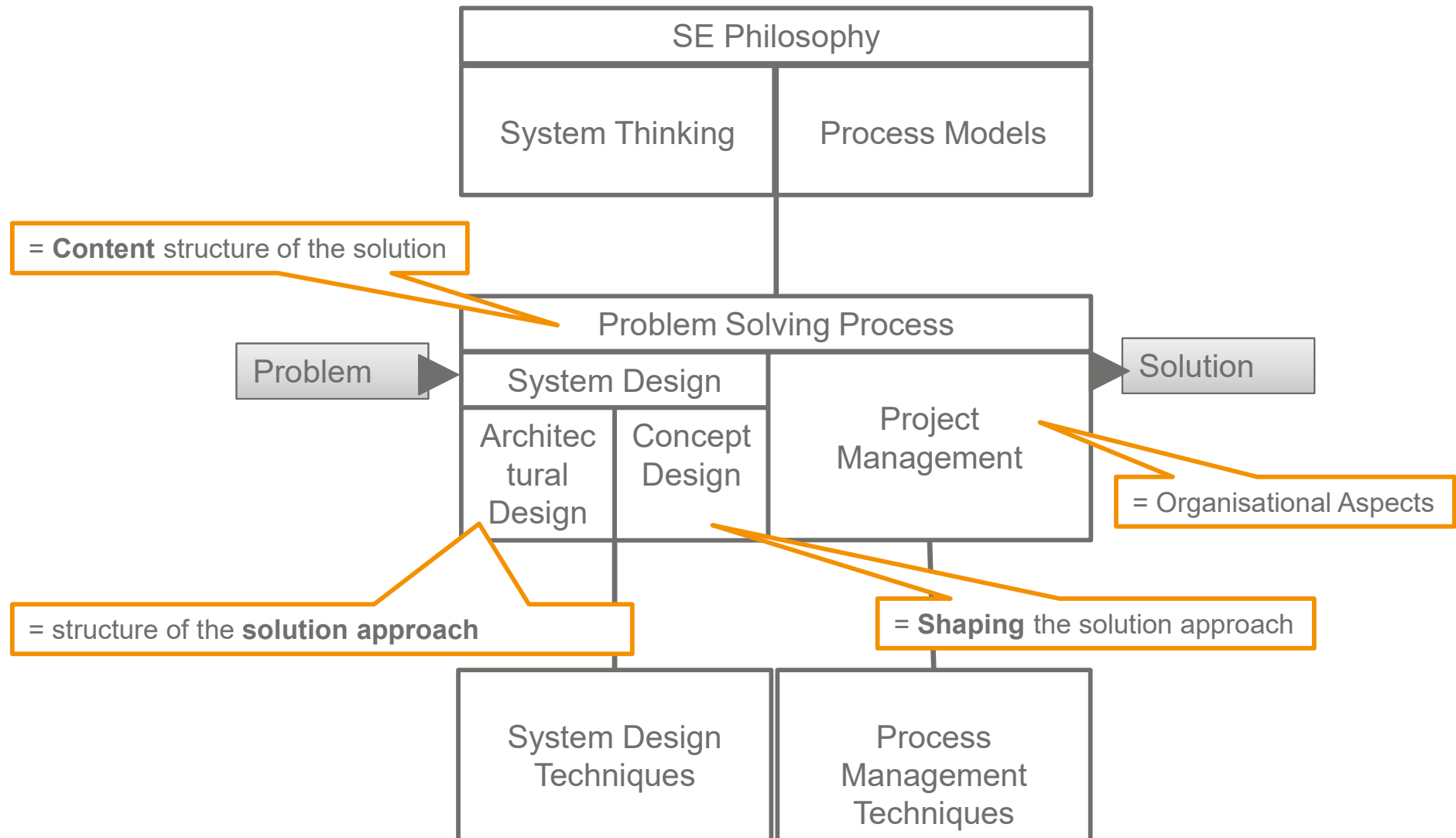
“ ... Systems Engineering as a “Bindestrich-Disziplin” of several disciplines...”

Recap:

What are important components in SE?

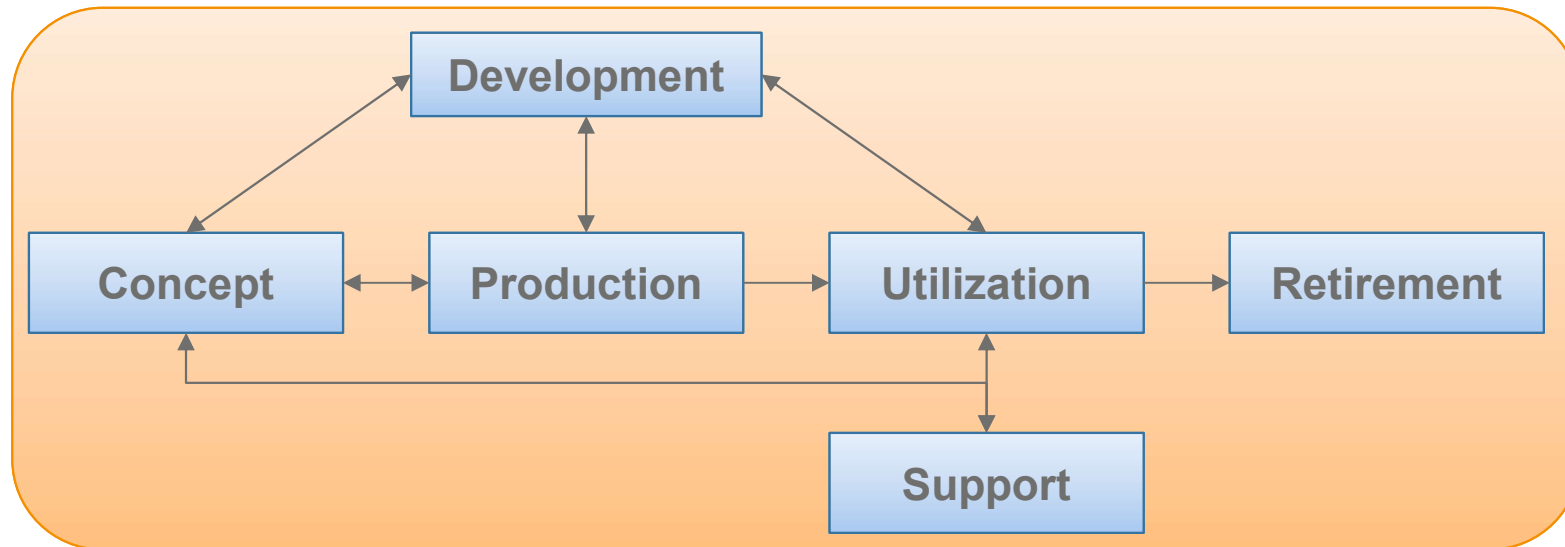
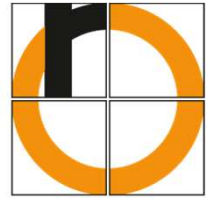


- System Engineering Concept



(1) Source: Haberfellner et al., Systems Engineering Grundlagen und Anwendung, orell füssli 2018 (14. Auflage)

Generic Product Life Cycle Model



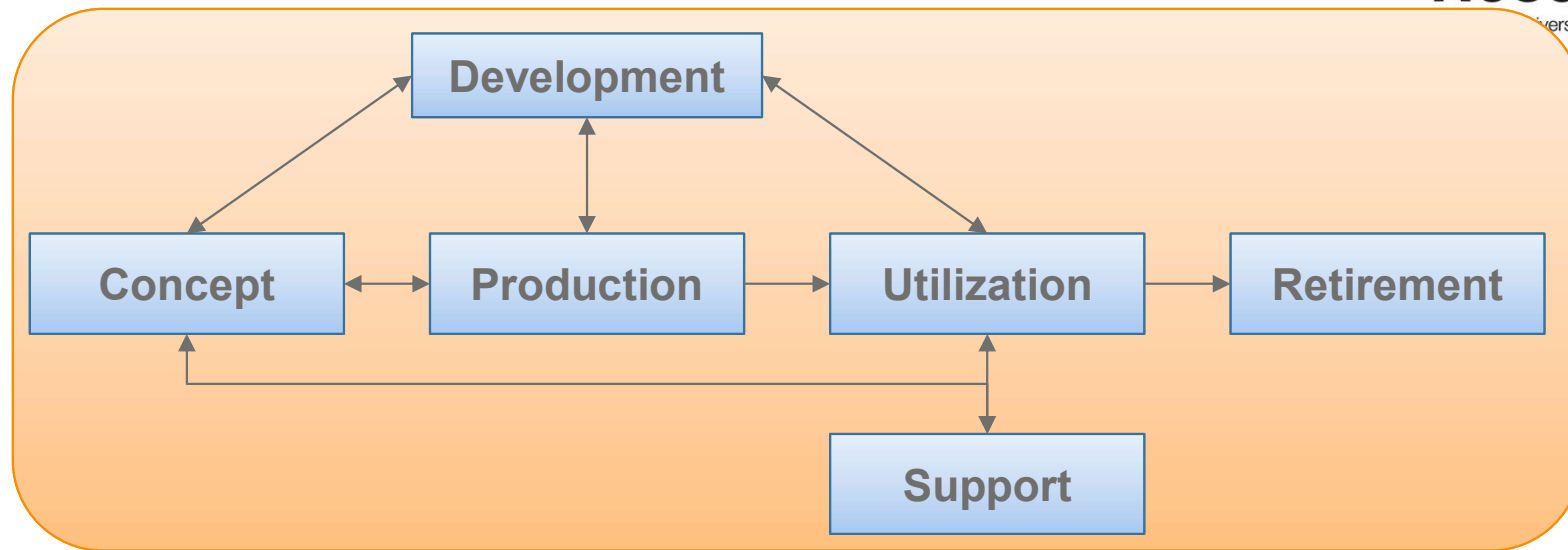
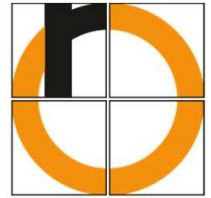
“ ... Life cycles vary according to the nature, purpose, use and prevailing circumstances of the system ...” (1, also graph)

“ ... Life cycles provide a framework for meeting the stakeholder/customer needs and to keep the project in time. ...” (1)

- **Efficient methods**
- **Life cycle stages**
- **Decision gates** to decide readiness of life cycle stages (see next slide)

(1) INCOSE, Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities, Wiley, 2015

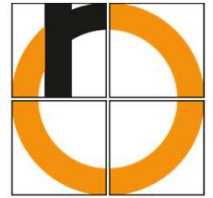
Generic Product Life Cycle Model



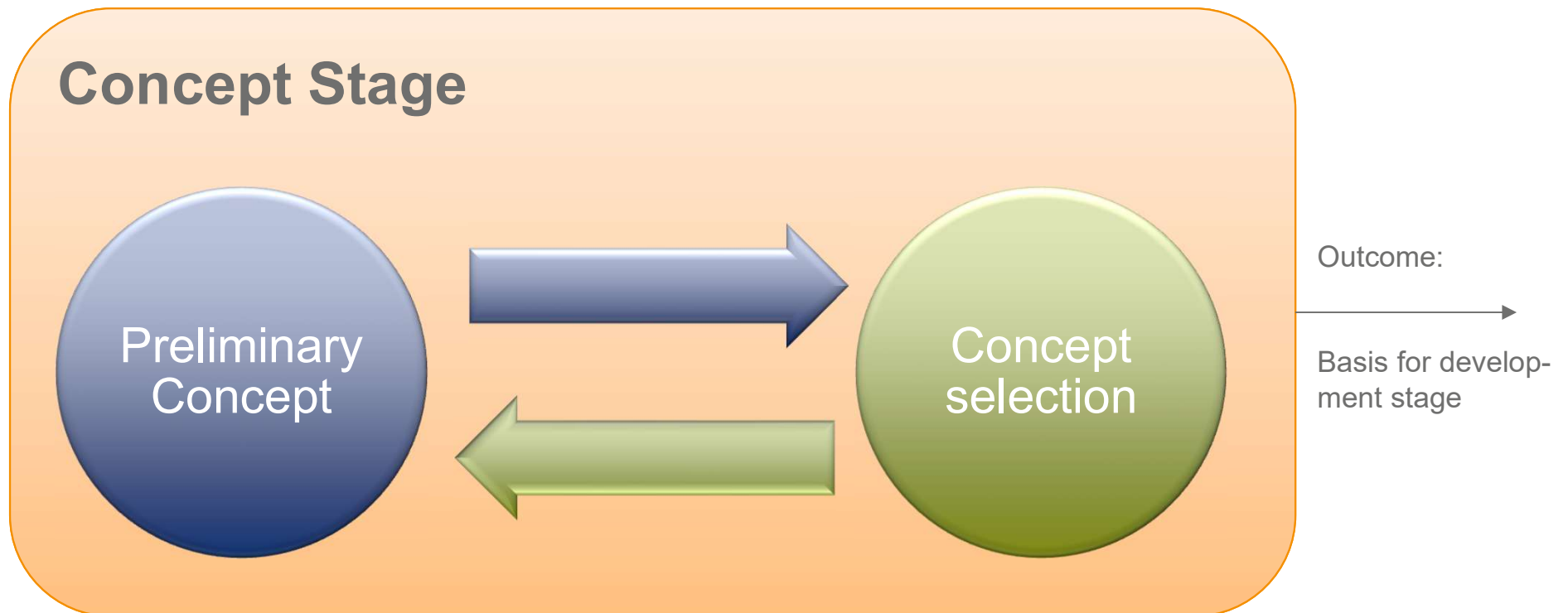
(1) INCOSE, Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities, Wiley, 2015

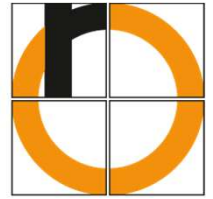
- It is most likely the combination of HW and physical processes AND SW to realize and control the system.
- System Engineer: helps building a technology that is aligned with business case & funding.
- Three system aspects:
 - Business case ↔ Funding ↔ Technology
 - System integrity: all three aspects are weighted equally important
- Systems Engineering is important in all life cycle stages! Why? Examples ...
 - Needs can change during production phase
 - Needs can change during operation
 - System technology evolves
 - Data from utilization phase can give valuable insights into new system versions (performance analysis, interface analysis, management, etc.)

Concept stage



- Create a comprehensive picture of the project to have a reliable risk, cost, and time prediction
- Concept stage requires creative System Engineering
- System Engineer: coordination





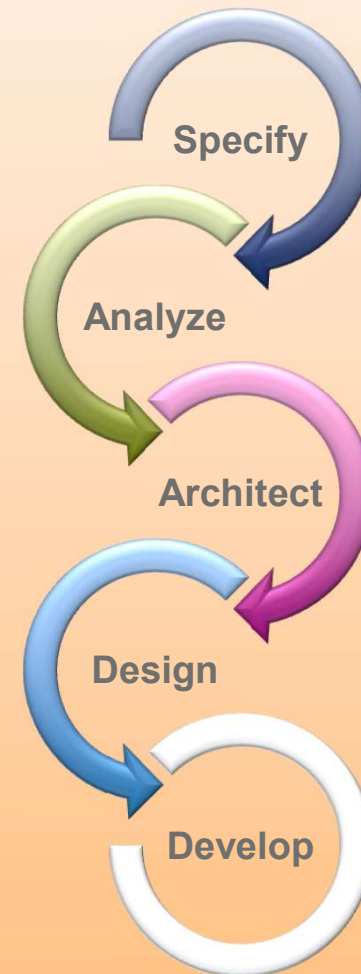
Development Stage

- Realization phase of the system to meet stakeholder requirements and to have a system that can be produced, utilized, supported, and retired.
- Basis: output of concept stage
- Outcome: system of interest (HW, SW, so the product), documentation, models, cost estimates, enabling systems, etc.

Basis:

Output of
concept stage

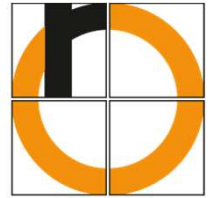
hardware, software



Outcome:

System of interest

Production and Utilization Stages



Production Stage:

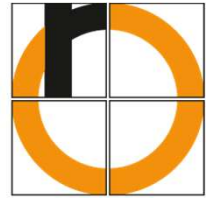
- System manufacturing
- Adaptations possible, i.e., if there are production problems, too high production costs, or to further improve system capabilities → system requirement adaption → re-validation & verification → undergoes Software Engineering assessment before approval

thus bi-directional to concept stage

Utilization Stage:

- Operation of the system in its supposed environment
- Product modifications are planned → enhance system within next cycle or through system integration at run-time
- Complex technical systems: midlife upgrades (HW & SW)

Support and Retirement Stages



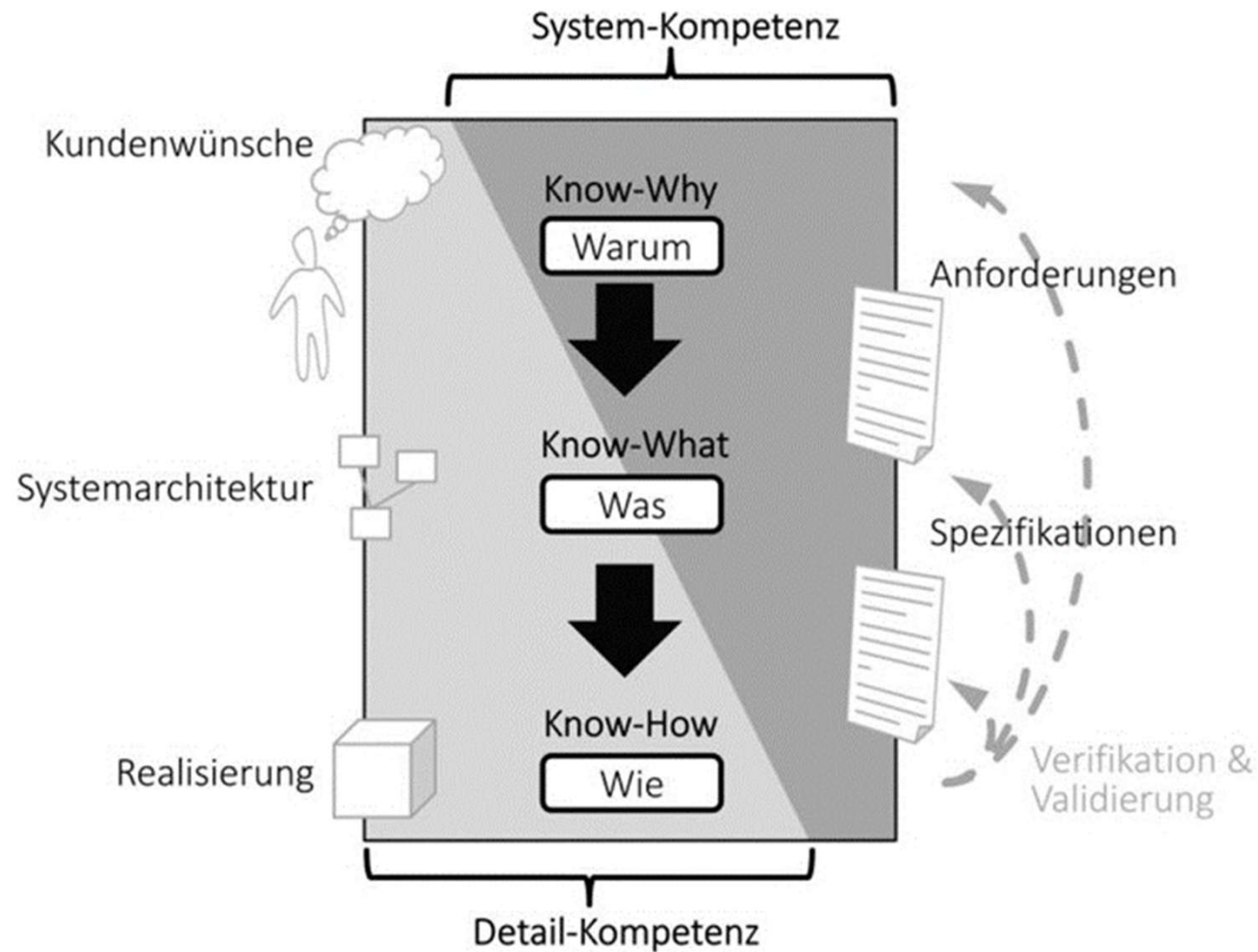
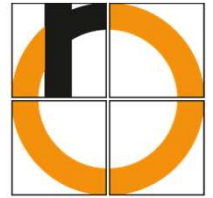
Support Stage:

- == Service
- System modification suggestions to enable better support, lower operational cost and to extend lifetime of system → undergoes Software Engineering assessment before approval

Retirement Stage:

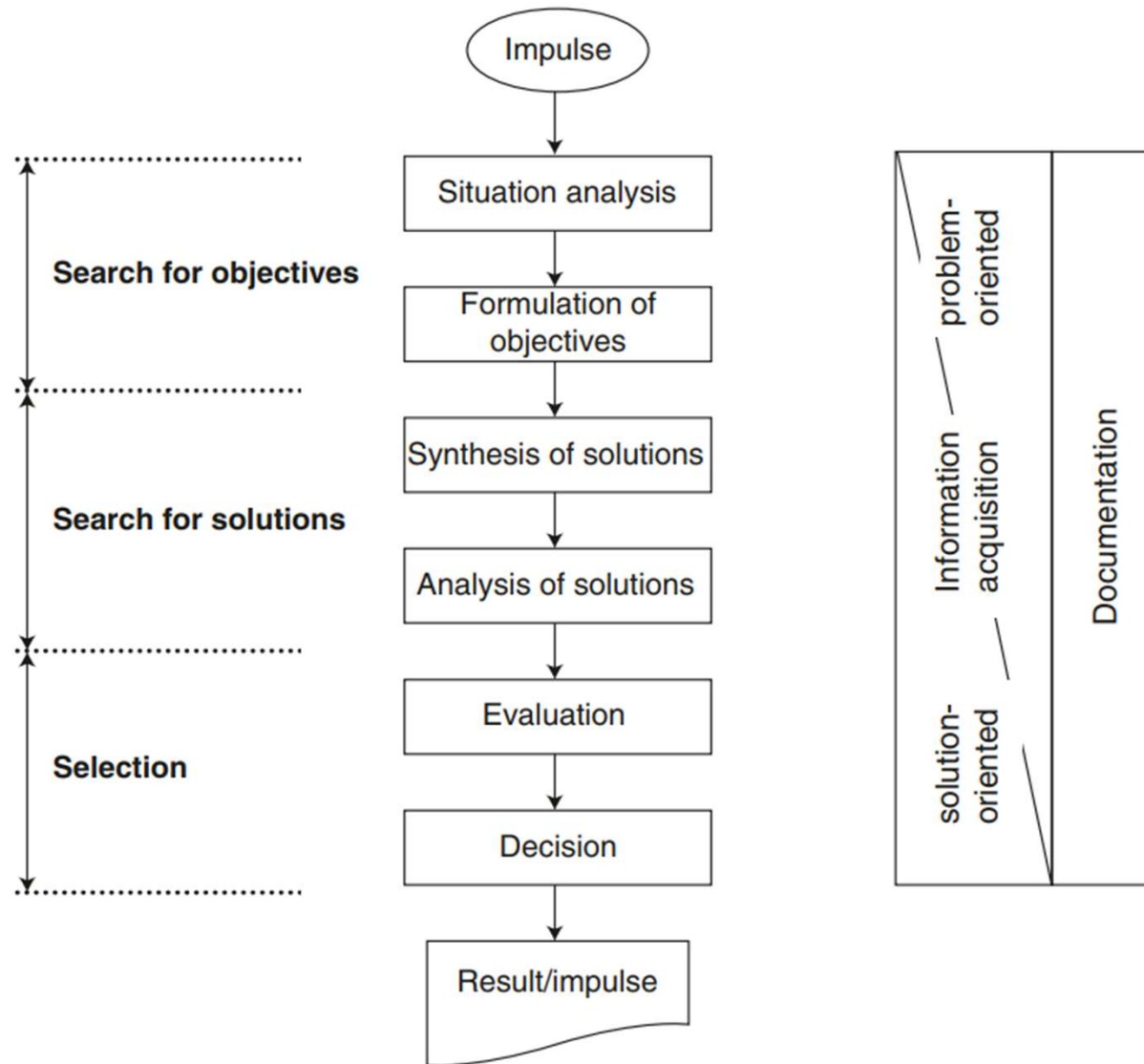
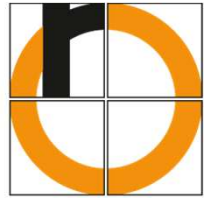
- System and all the involved elements (i.e. services) are removed from operation
- Systems Engineering mainly focuses that system disposal requirements are met
- Attention: system disposal is planned during concept stage; typically, system developer is responsible for proper end-of-life disposal.

SE Processes: Basic Principle



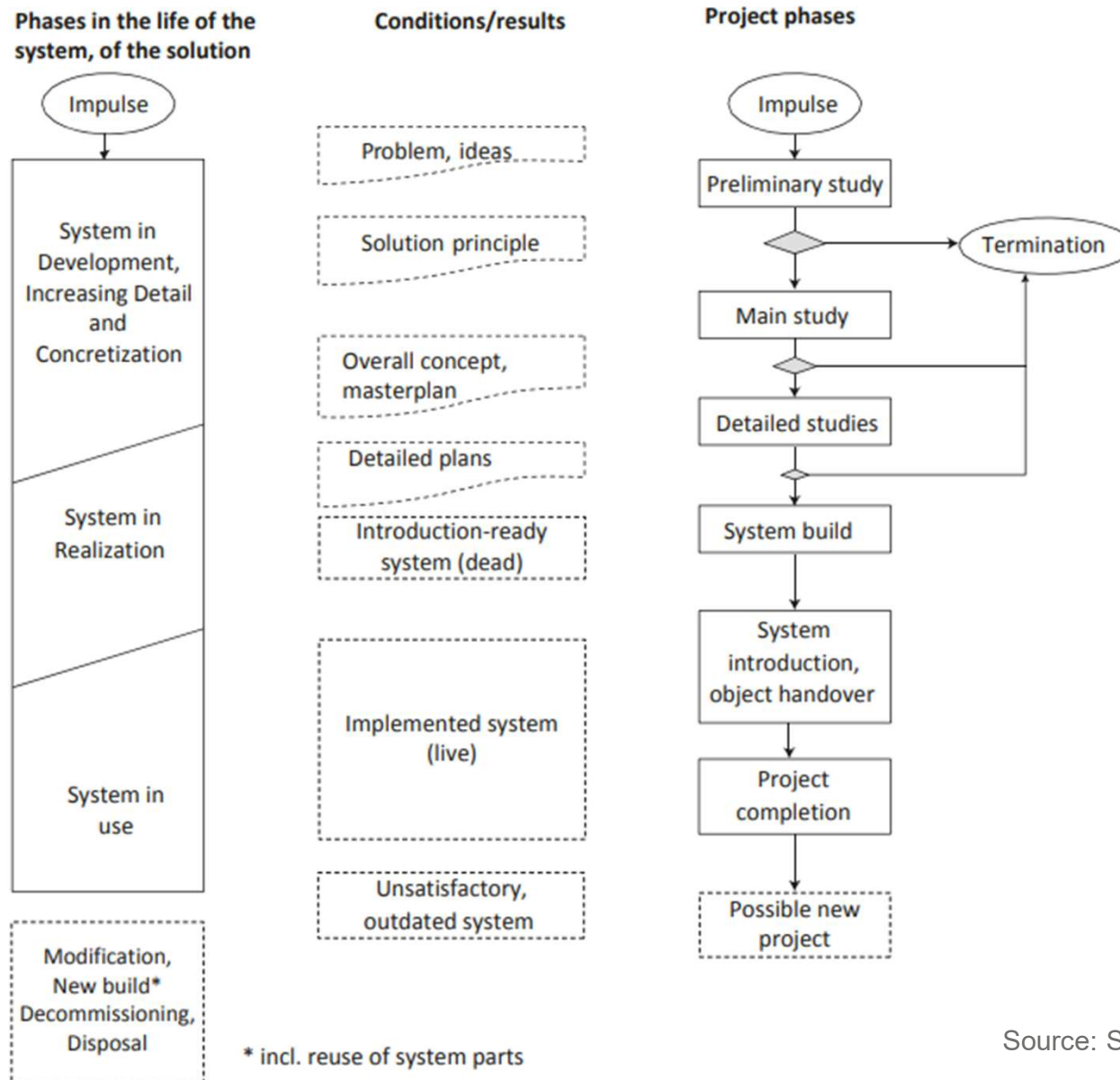
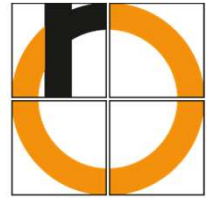
Source: GfSE: Figure 18, page 77: Grundmuster des SYSE

SE Processes: Finding / Making Decisions



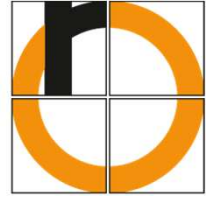
Source: SE (en): Figure 2.6, page 44: Problem-solving cycle (PSC) – basic version

SE Processes: Finding / Making Decisions



Source: SE (en): Figure 3.5, page 112: Phase concept – basic version

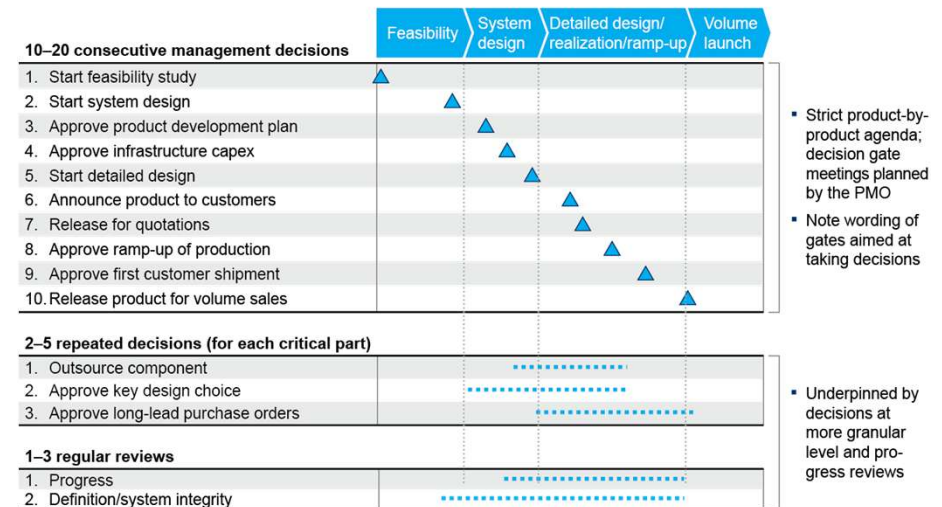
Decision Gates



- == Milestones (not every milestone in a product development is a decision gate, but every decision gate is a milestone)
- A decision gate is a mechanism to decide whether the defined deliverables of the system are satisfactorily achieved and that there is no risk in proceeding the next deliverable.

→ risk reduction, guarantee a successful implementation of the next deliverable

- Is the project on the right track → Deliverables satisfy business case?
- Is the project still affordable?
- Is the project still in time?
- Note: frequent interaction with customer/stakeholder may reduce (not eliminate) the amount of decision gates (Agile development).



<https://www.mckinsey.com/>

Life Cycle Stages

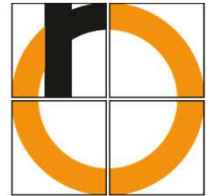
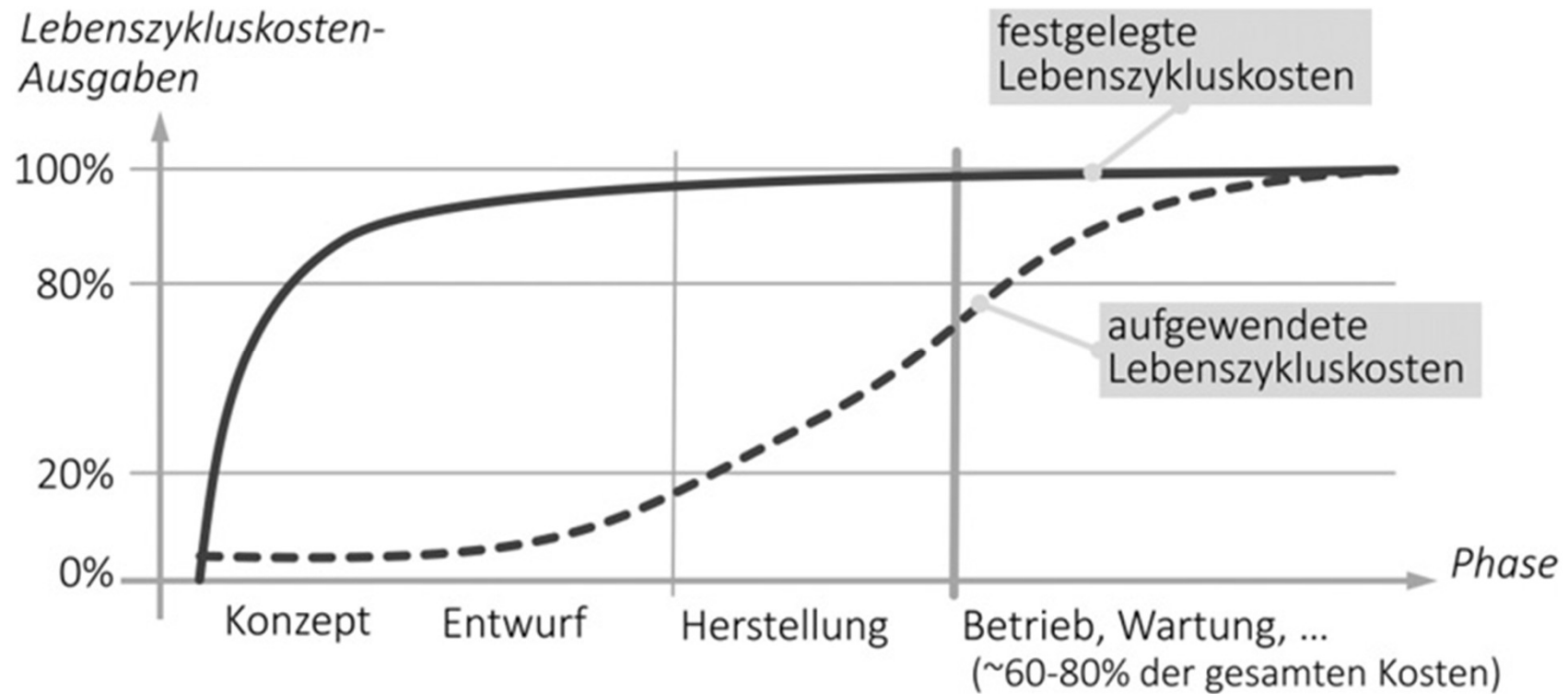
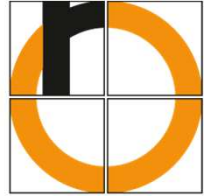


TABLE 3.1 Generic life cycle stages, their purposes, and decision gate options

Life cycle stages	Purpose	Decision gates
Concept	Define problem space 1. Exploratory research 2. Concept selection Characterize solution space Identify stakeholders' needs Explore ideas and technologies Refine stakeholders' needs Explore feasible concepts Propose viable solutions	Decision options • Proceed with next stage • Proceed and respond to action items • Continue this stage • Return to preceding stage • Put a hold on project activity • Terminate project
Development	Define/refine system requirements Create solution description—architecture and design Implement initial system Integrate, verify, and validate system	
Production	Produce systems Inspect and verify	
Utilization	Operate system to satisfy users' needs	
Support	Provide sustained system capability	
Retirement	Store, archive, or dispose of the system	

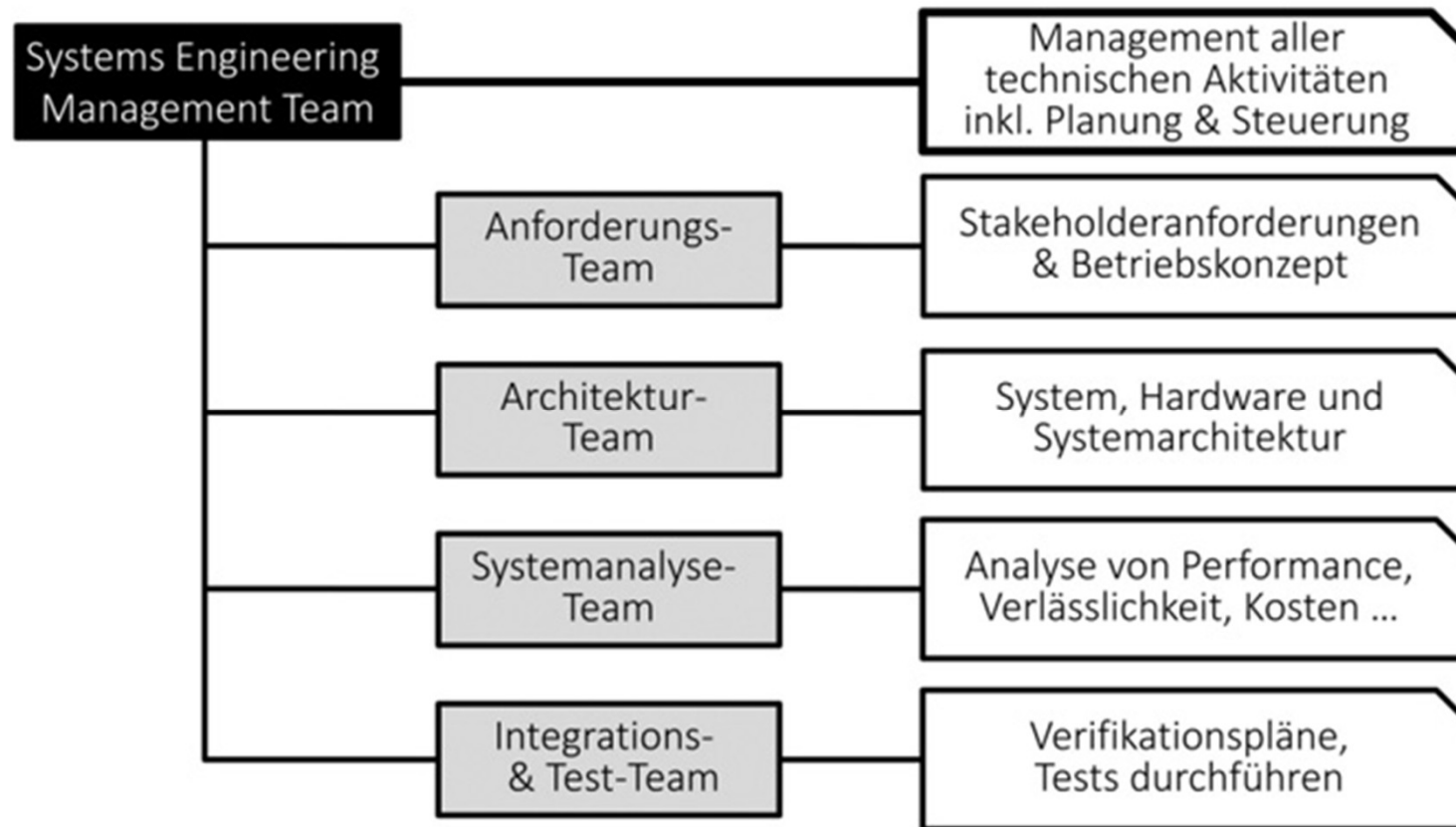
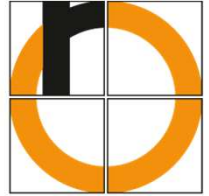
Table taken from (1), p. 28

Estimated vs. real cost

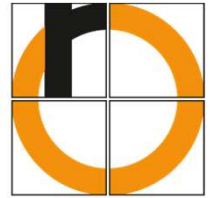


Source: GfSE: Figure 20, Estimated vs. real cost

SE Team Organization

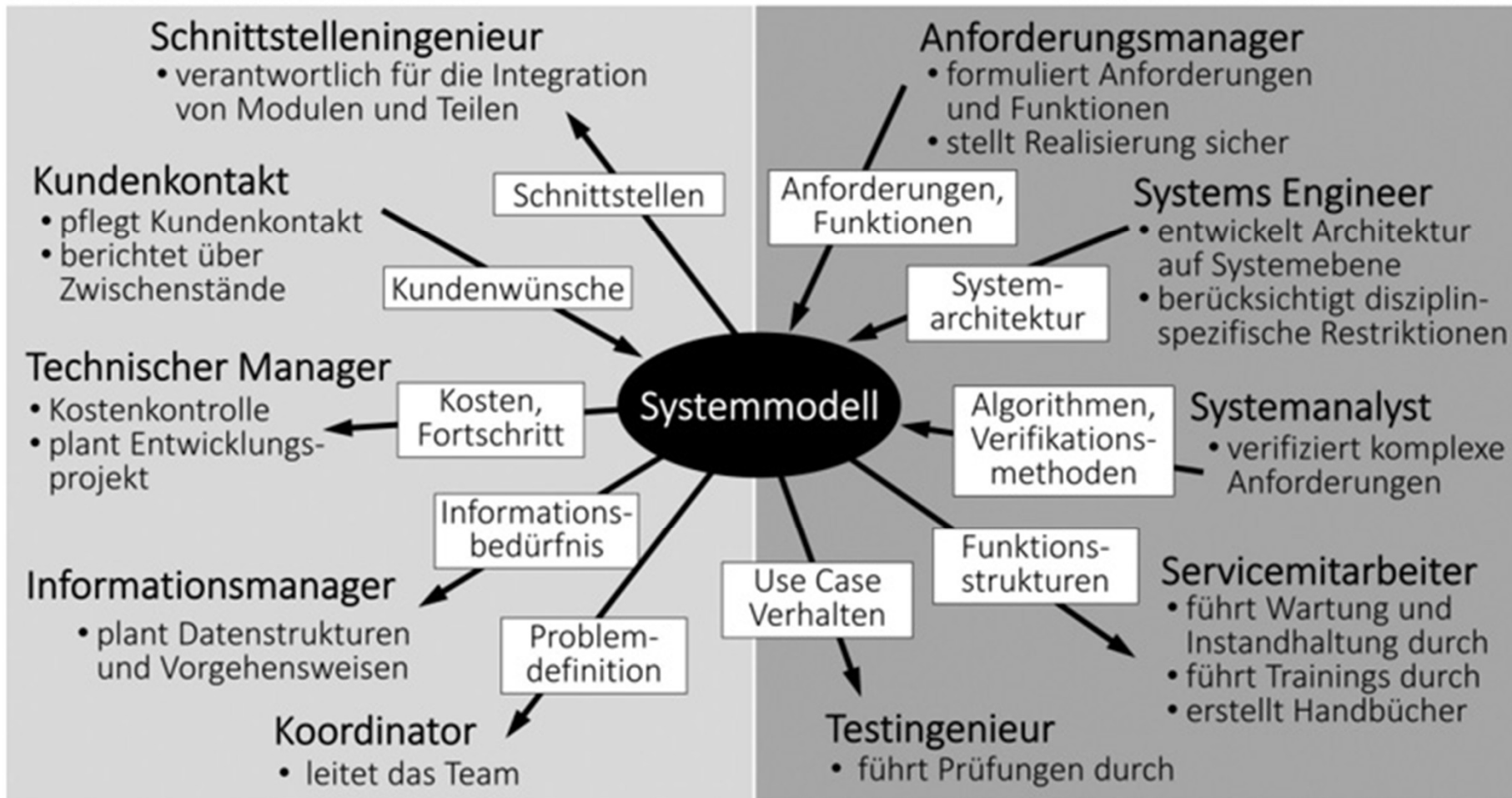


Source: GfSE: Figure 11, page 61: SYSE Team nach FRIEDENTHAL



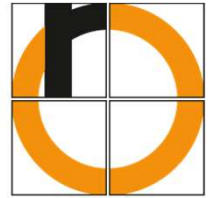
Projektmanagement-Rollen

Projektlebenszyklus-Rollen

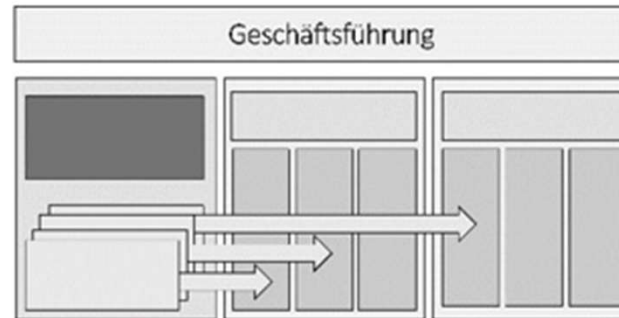


Source: GfSE: Figure 12, page 62: SYSE roles nach SHEARD

SE Organization Forms

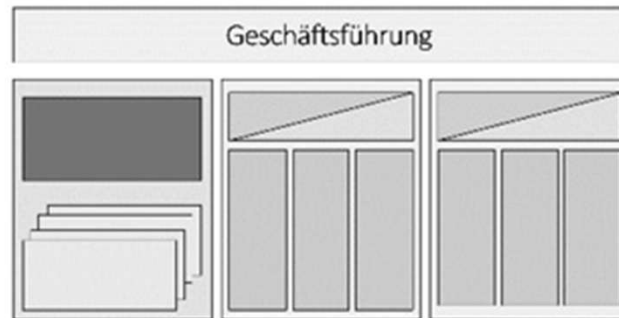


Systems Engineering als Service



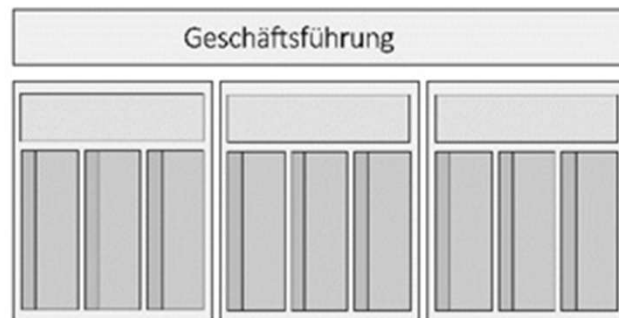
SE als eine zentrale
Serviceeinheit
→ Projektübergreifend

Projektbezogenes Systems Engineering und zentraler Support



SE projektbezogen & zentrale
Einheit

Systems Engineering als Linienorganisation

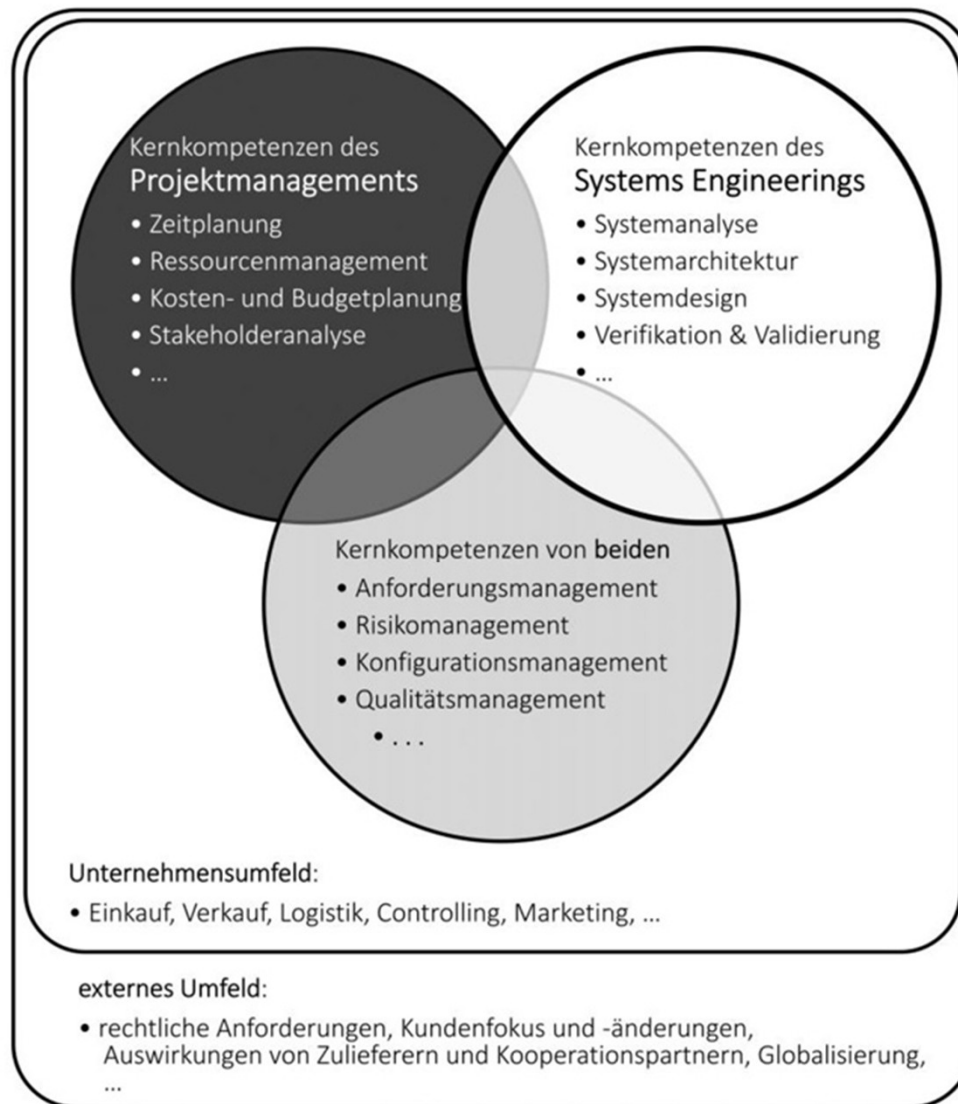
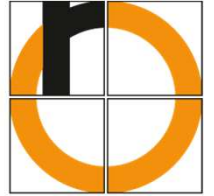


SE fest in Abteilung verankert

jedes Abteil hat eigene System Engineers

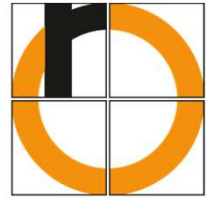
Source: GfSE: Figure 13, page 63: SYSE organisation forms

Interaction of SE and PM



Source: GfSE: Figure 10, page 50: Interaction of SE and PM

Current and Future Trends in SE



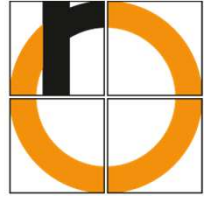
Current

- Standalone systems
- Relatively stable requirements
- Requirements determine skills
- Control over evolution of custom systems
- Failures locally critical
- Completely defined systems with specific functionality
- Repeatability-oriented process, maturity models
- Enough time to stay stable
- Stable jobs

Future

- Everything connected (maybe)
- Rapid requirement changes
- Commercial off-the-shelf (COTS) capabilities determine requirements
- No control over evolution COTS products
- Ever-decreasing cycle times
- Outsourced jobs
- System failures broadly critical
- Complex, adaptive, emergent systems of systems
- Adaptive process models

System Live Cycle Models



Plan-driven

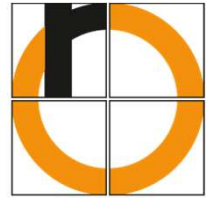
- Waterfall model
- Waterfall model with iterations
- V-model
- V-model XT
- SIMILAR

Agile

- Incremental Life Cycle
- Spiral
- Agile MBSE

je iterativer, desto leichter ist es agil zu arbeiten > schnellere Evaluierung des Feedbacks, schnellere Reaktionen

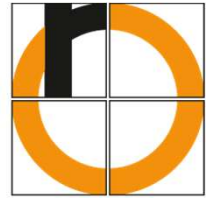
Plan-Driven Sequential System Life Cycle Models



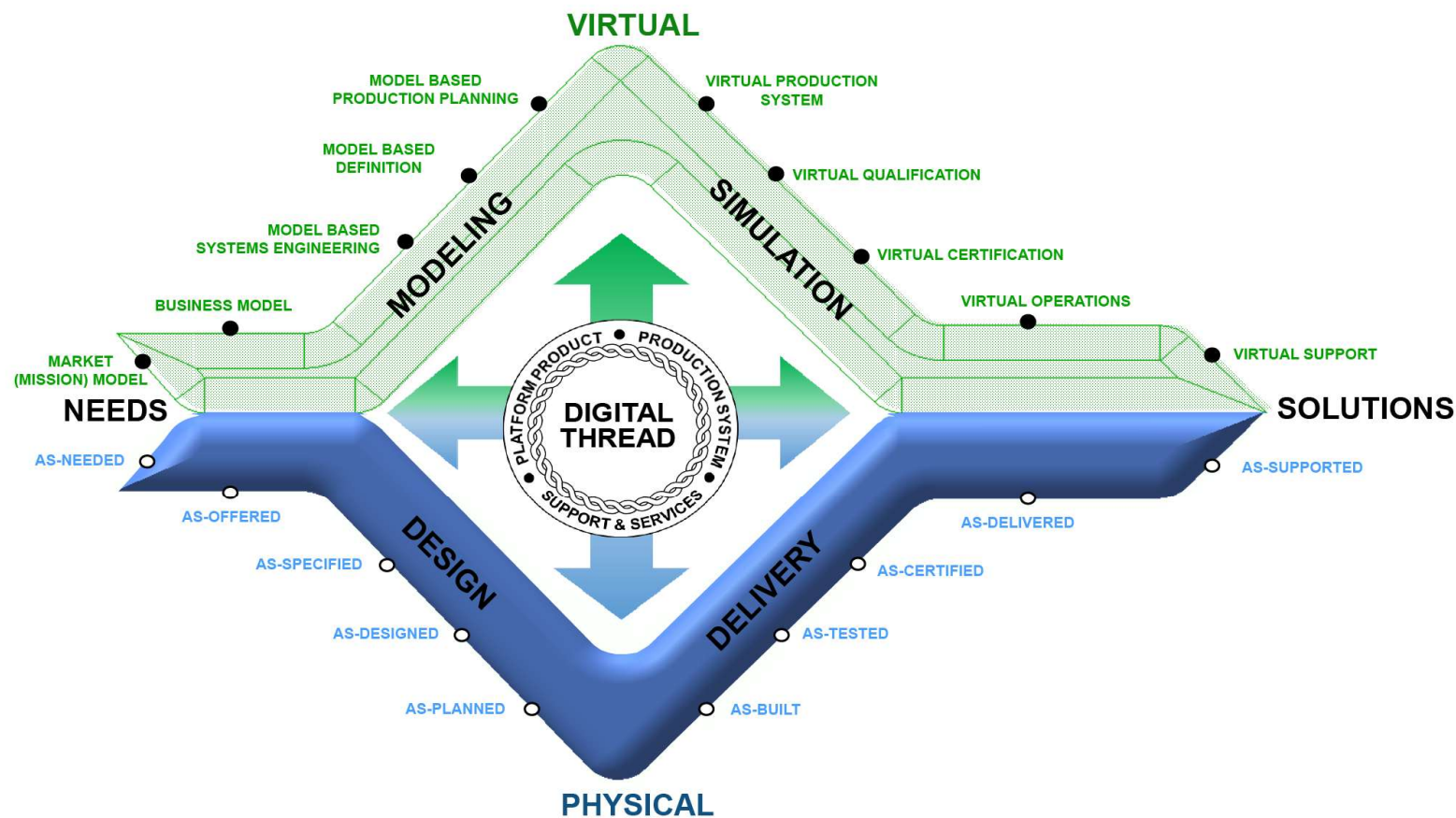
- Good for projects with a large team and several companies involved
- Systematic approach → fulfillment to predefined, fixed processes within the life cycle stages → master plan
- Good communication mechanism between different involved groups, e.g. developers, business level, etc.
- All the knowledge already accumulated in the form of data, information and expertise can be used as a prerequisite for future planning.
- Focus: good documentation, high traceability of requirements, every system status is verified
- Benefits:
 - Predictability
 - Stability
 - Repeatability
 - High assurance
- Drawbacks:
 - Potentially higher cost in case of changes
 - Linear, iterative and recursive properties of involved tasks are hidden.
- Example: Vee-Model, Waterfall Model

Example

Diamond : virtual flipped V-model

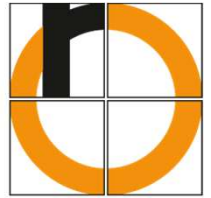


Evolution of System Engineering (SE) to Model Based Engineering (MBE)



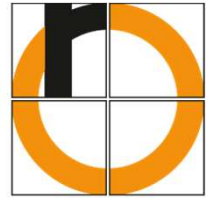
Source: <https://www.boeing.com/modelbasedengineering.html> (BOEING)

Incremental and Iterative System Life Cycle Methods (IID)



- Goal: rapid value and responsiveness
- Good in case requirements are unclear from beginning or if system should be open in terms of adding new technologies into the system → enable flexibility
- A candidate system of interest is developed and assessed – does it meet stakeholders requirements/needs? NO: initiate another round of developing a candidate system until a YES is reached.
- Good for smaller teams, or to system elements
- Stakeholders are included into the working stages → early feedback
- Benefits:
 - Flexible
 - Events can be taken out of sequence (in case of low risk)
 - Velocity – speed and direction
 - Adaptable
 - Avoids large investments
- Drawbacks:
 - Reactive project (i.e. stakeholder opinion changes) – can be chaotic
- Example: Spiral Model, evolutionary development (i.e. in R&D)

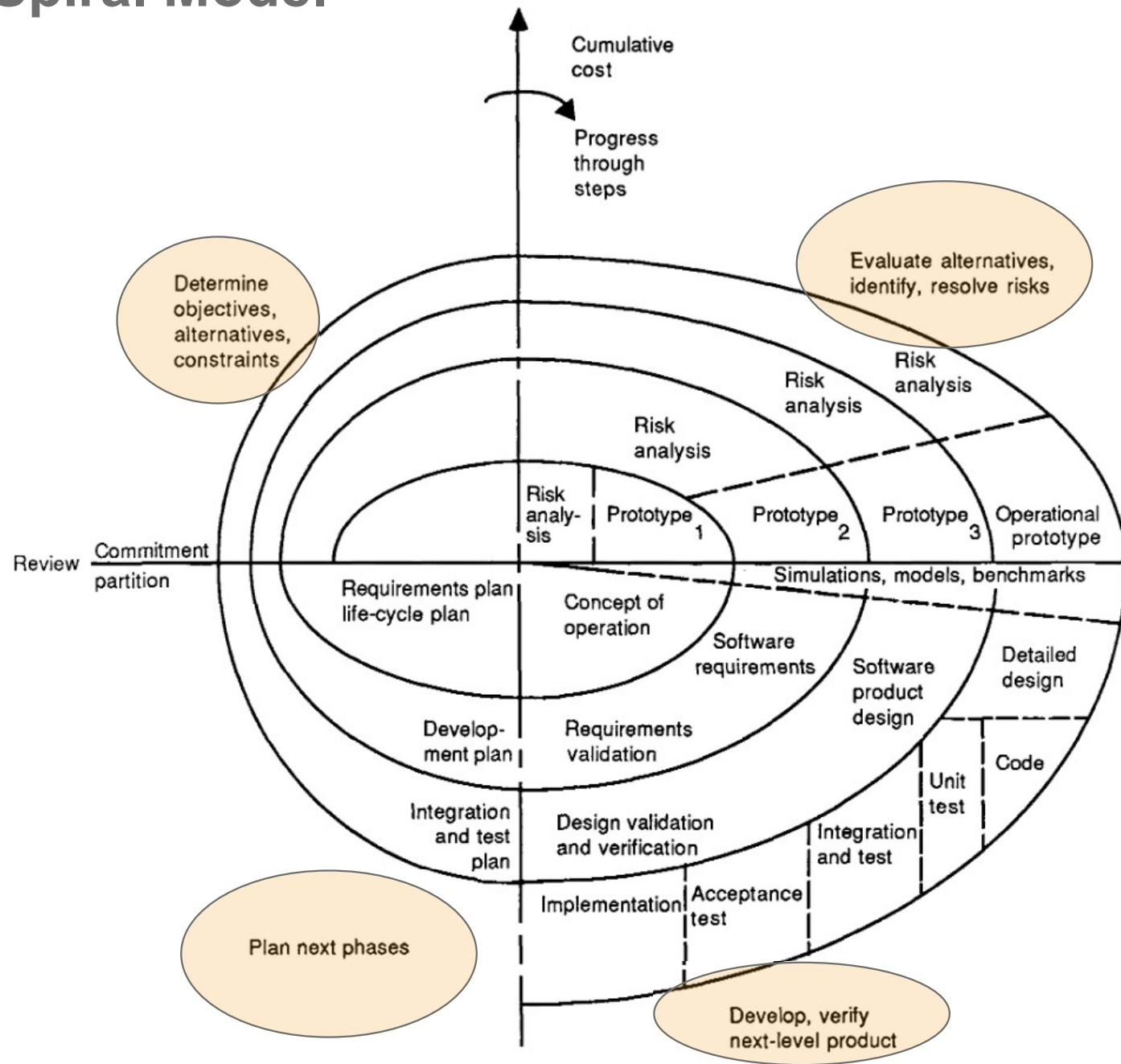
Spiral Model



detailreicher je weiter außen in der spirale

output einer stage ist input der nächsten

decision gates bestimmen übergang in die nächste stage



Each iteration
passes all
phases 1 to 4

Graph taken from (4)

Time for questions ...

